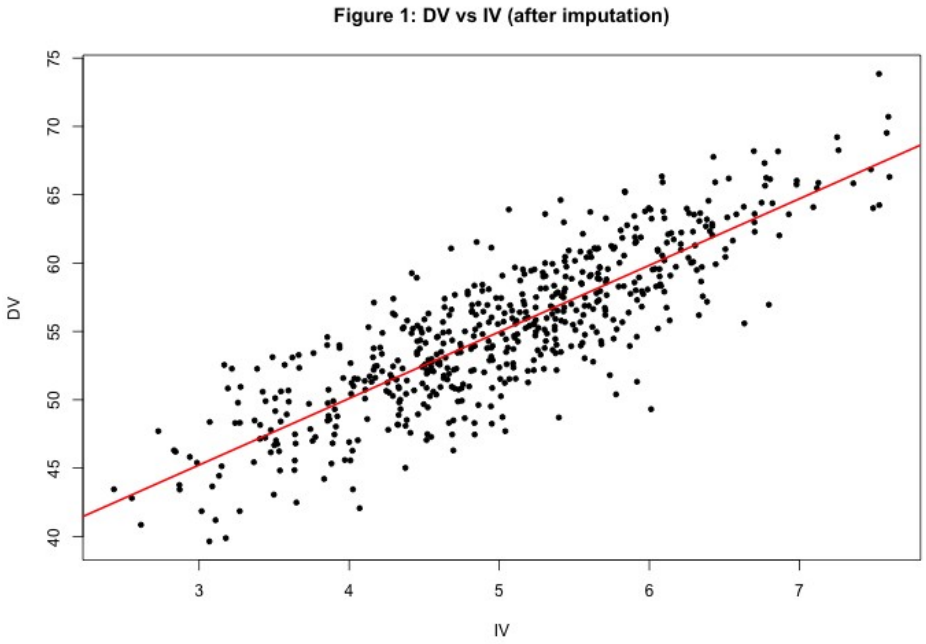
AMS 315: Computing Assignment 1 – Appendix

Kyle Chan

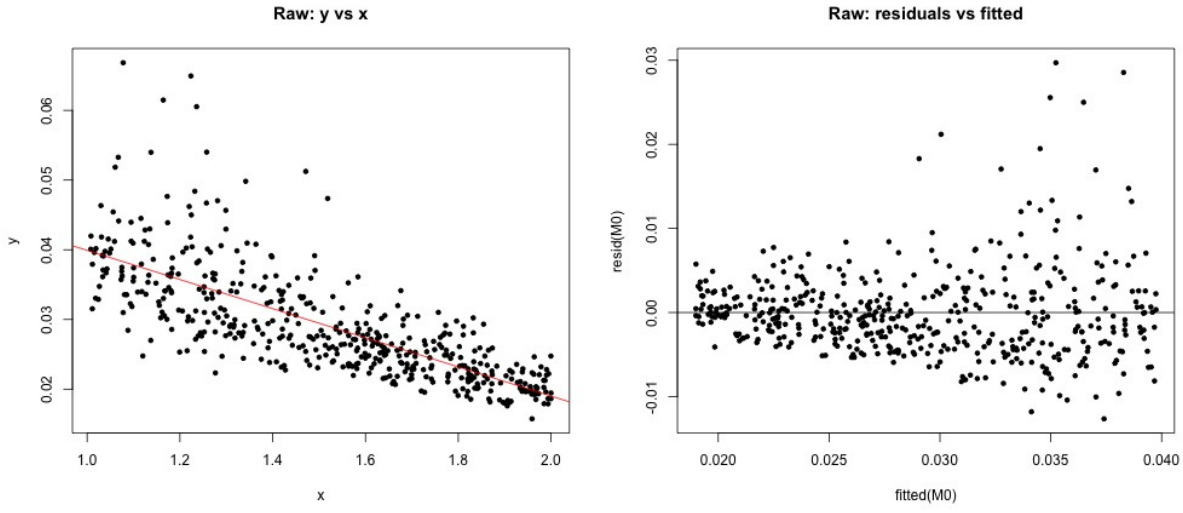
Appendix A: Figures

Figure 1. Scatterplot of DV vs. IV with fitted regression line (Part A, after imputation)



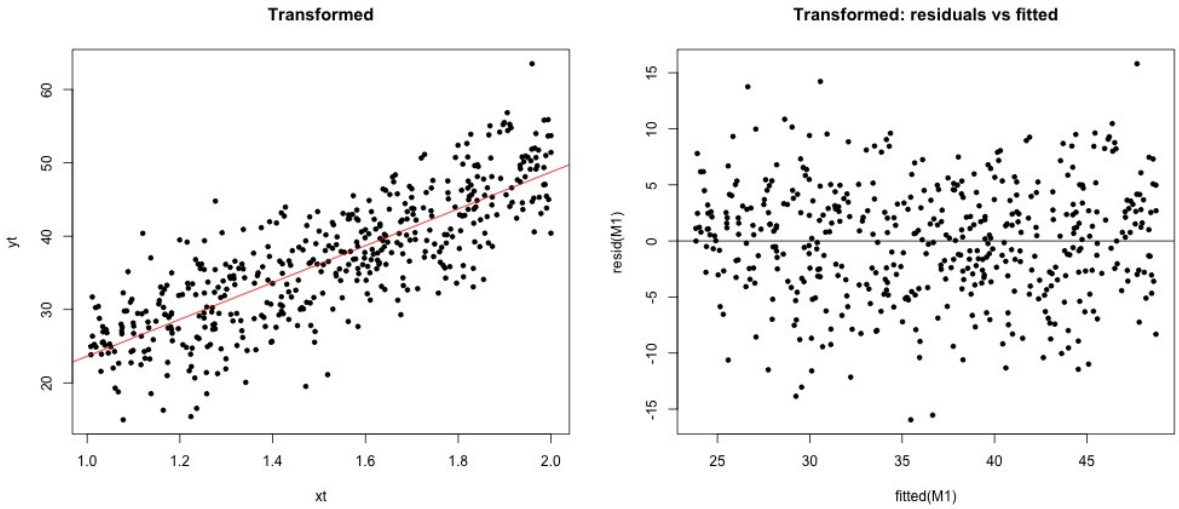
The red line is the fitted function DV = 30.62 + 4.87·IV. The points follow a clear positive linear trend with fairly even scatter around the line.

Figure 2. Raw data (Part B): scatterplot of y vs. x (left) and residuals vs. fitted values (right)



The scatterplot curves downward, and the residuals fan out as the fitted values increase (a funnel shape), so the constant-variance assumption is not met.

Figure 3. Transformed data (Part B): scatterplot of 1/y vs. x (left) and residuals vs. fitted values (right)



After the 1/y transformation, the relationship is close to linear and the residuals form a random band with roughly constant spread.

Appendix B: Tables

Table 1. Subject ID counts (Part A)

Category	Count
Total subject IDs	569
IDs with an IV value	500
IDs with a DV value	520
IDs with both IV and DV	454
IDs with at least one of IV or DV	566
IDs missing both	3
Fraction of IV missing	0.1213 (12.1%)
Fraction of DV missing	0.0861 (8.6%)

Table 2. ANOVA for the regression of DV on IV (Part A)

Source	df	Sum Sq	Mean Sq	F	p-value
IV	1	13020.3	13020.3	1428.9	< 2.2e-16
Residuals	564	5139.4	9.1		

Table 3. R^2 for candidate transformations (Part B; rows = x transform, columns = y transform)

	y	**log(y)**	**1/y**	**√y**	**1/√y**
x	0.5610	0.6330	**0.6602**	0.6026	0.6520
log(x)	0.5626	0.6260	0.6434	0.6002	0.6401
1/x	0.5547	0.6087	0.6167	0.5877	0.6180
√x	0.5630	0.6308	0.6530	0.6026	0.6473
x^2	0.5505	0.6302	0.6668	0.5956	0.6538

Bold: the chosen model (1/y vs. x).

Table 4. ANOVA for 1/y regressed on x (Part B)

Source	df	Sum Sq	Mean Sq	F	p-value
x	1	23918	23917.7	916.98	< 2.2e-16
Residuals	472	12311	26.1		

Table 5. Lack-of-fit test, 1/y on binned x (bin width 0.05, 19 bins) (Part B)

Source	df	Sum Sq	Mean Sq	F	p-value
x (binned)	1	23845.6	23845.6	919.98	< 2e-16
Residuals	472	12383.3	26.2		
Lack of fit	17	589.8	34.7	1.339	0.164
Pure error	455	11793.5	25.9		

Appendix C: R Code

File: run_all.R

```
# Runs the full analysis and saves all printed output to output/
dir.create("output", showWarnings = FALSE)

sink("output/partA_output.txt", split = TRUE)
source("R/partA.R", echo = TRUE)
sink()

sink("output/partB_output.txt", split = TRUE)
source("R/partB.R", echo = TRUE)
sink()
```

File: R/partA.R

```
# ============================================================
# AMS 315 - Project 1, Part A
# Merge IV + DV files by subject ID, count missing data, impute
# Run from the project folder (open ams315-project1.Rproj or setwd() to it)
# ============================================================

# install.packages("mice")   # run once
library(mice)

set.seed(315)   # makes the imputation reproducible

# ---- 1. Read the two files ---------------------------------
iv <- read.csv("data/partA_IV.csv", header = TRUE)
dv <- read.csv("data/partA_DV.csv", header = TRUE)
str(iv); str(dv)

# ---- 2. Sort by ID and merge (NOT cut-and-paste) ------------
# all = TRUE is a FULL outer join: keeps an ID even if it only
# appears in one file, so no subjects are silently dropped.
iv <- iv[order(iv$ID), ]
dv <- dv[order(dv$ID), ]
partA <- merge(iv, dv, by = "ID", all = TRUE)
str(partA)
write.csv(partA, "data/partA_merged.csv", row.names = FALSE)

# ---- 3. Counts the report asks for -------------------------
n_total   <- nrow(partA)
has_iv    <- !is.na(partA$IV)
has_dv    <- !is.na(partA$DV)

counts <- data.frame(
  Quantity = c("Total subject IDs",
               "IDs with an IV value",
               "IDs with a DV value",
               "IDs with both IV and DV",
               "IDs with at least one of IV or DV",
               "IDs missing both",
               "Fraction of IV missing",
               "Fraction of DV missing"),
  Value = c(n_total,
            sum(has_iv),
            sum(has_dv),
            sum(has_iv & has_dv),
            sum(has_iv | has_dv),
            sum(!has_iv & !has_dv),
            round(mean(!has_iv), 4),
            round(mean(!has_dv), 4))
)
print(counts)
md.pattern(partA)   # visual missing-data pattern (good for the appendix)

# ---- 4. Drop rows with NO information, then impute ----------
partA_imp <- partA[has_iv | has_dv, ]
```

```
# norm.boot = linear regression imputation with bootstrap.
# Other allowed choices: "pmm" (predictive mean matching), "norm".
# NOT allowed: mean, median, listwise deletion.
imp <- mice(partA_imp[, c("IV", "DV")], method = "norm.boot",
            m = 5, printFlag = FALSE)
partA_complete <- cbind(ID = partA_imp$ID, complete(imp))
md.pattern(partA_complete)   # should say "completely observed"
write.csv(partA_complete, "data/partA_imputed.csv", row.names = FALSE)

# ---- 5. Regression on the completed data --------------------
M <- lm(DV ~ IV, data = partA_complete)
summary(M)
anova(M)
confint(M, level = 0.95)

png("figures/partA_scatter.png", width = 700, height = 500)
plot(DV ~ IV, data = partA_complete, pch = 20,
     main = "Figure 1: DV vs IV (after imputation)")
abline(M, col = "red", lwd = 2)
dev.off()
```

File: R/partB.R

```
# ============================================================
# AMS 315 - Project 1, Part B
# Find a transformation, fit regression, bin x, lack-of-fit test
# ============================================================

# Run once to install alr3 (provides pureErrorAnova):
# install.packages("remotes")
# remotes::install_github("cran/alr3")   # if asked to update packages, type 3
library(alr3)

dat <- read.csv("data/partB.csv", header = TRUE)
str(dat)
summary(dat)

# ---- 1. Look at the raw data --------------------------------
M0 <- lm(y ~ x, data = dat)
summary(M0)$r.squared            # baseline R-squared

png("figures/partB_raw.png", width = 1000, height = 450)
par(mfrow = c(1, 2))
plot(y ~ x, data = dat, pch = 20, main = "Raw: y vs x"); abline(M0, col = "red")
plot(fitted(M0), resid(M0), pch = 20, main = "Raw: residuals vs fitted"); abline(h = 0)
dev.off()

# ---- 2. Try candidate transformations (textbook 11.1) --------
# Compare R-squared for each combination. Also LOOK at each
# residual plot -- the best choice is the one whose residuals
# look patternless with roughly constant spread.
tx <- list(x = function(v) v, `log(x)` = log, `1/x` = function(v) 1/v,
           `sqrt(x)` = sqrt, `x^2` = function(v) v^2)
ty <- list(y = function(v) v, `log(y)` = log, `1/y` = function(v) 1/v,
           `sqrt(y)` = sqrt, `1/sqrt(y)` = function(v) 1/sqrt(v))
r2 <- outer(names(tx), names(ty), Vectorize(function(a, b) {
  summary(lm(ty[[b]](dat$y) ~ tx[[a]](dat$x)))$r.squared
}))
dimnames(r2) <- list(names(tx), names(ty))
round(r2, 4)

# Box-Cox on y can suggest a power (lambda): -1 = 1/y, 0 = log(y), 0.5 = sqrt(y)
# library(MASS); boxcox(M0)

# ---- 3. Fit your chosen model -------------------------------
# >>> EDIT these two lines to the transformation you picked <<<
dat$xt <- dat$x
dat$yt <- 1 / dat$y

M1 <- lm(yt ~ xt, data = dat)
```

```
summary(M1)
anova(M1)
confint(M1, level = 0.95)

png("figures/partB_transformed.png", width = 1000, height = 450)
par(mfrow = c(1, 2))
plot(yt ~ xt, data = dat, pch = 20, main = "Transformed"); abline(M1, col = "red")
plot(fitted(M1), resid(M1), pch = 20, main = "Transformed: residuals vs fitted"); abline(h = 0)
dev.off()

# ---- 4. Bin near-repeated x values ---------------------------
# Cut x into intervals; every x in an interval is replaced by the
# interval's mean x. Pick a width so each bin has a decent number
# of points (check table(groups)).
width  <- 0.05
groups <- cut(dat$xt,
              breaks = c(-Inf,
                         seq(min(dat$xt) + width, max(dat$xt) - width, by = width),
                         Inf))
table(groups)
dat$x_bin <- ave(dat$xt, groups)       # mean x within each bin

# ---- 5. Lack-of-fit test -------------------------------------
fit_bin <- lm(yt ~ x_bin, data = dat)
pureErrorAnova(fit_bin)
# Large p-value on the "Lack of fit" row => no evidence the straight
# line is the wrong shape for the transformed data.

# Optional: run the same LOF on the UNtransformed data for comparison
dat$x_bin_raw <- ave(dat$x, cut(dat$x, breaks = c(-Inf,
                     seq(min(dat$x) + width, max(dat$x) - width, by = width), Inf)))
pureErrorAnova(lm(y ~ x_bin_raw, data = dat))
```

Appendix D: R Output

File: output/partA_output.txt

```
> # ============================================================
> # AMS 315 - Project 1, Part A
> # Merge IV + DV files by subject ID, count missing .... [TRUNCATED]

> set.seed(315)   # makes the imputation reproducible

> # ---- 1. Read the two files ---------------------------------
> iv <- read.csv("data/partA_IV.csv", header = TRUE)

> dv <- read.csv("data/partA_DV.csv", header = TRUE)

> str(iv); str(dv)
'data.frame':   569 obs. of  2 variables:
 $ ID: int  440 467 313 446 44 374 201 258 525 45 ...
 $ IV: num  4.63 5.24 4.29 4.77 5.57 ...
'data.frame':   569 obs. of  2 variables:
 $ ID: int  332 368 164 230 347 323 313 287 269 547 ...
 $ DV: num  58.3 47.6 NA 59 58.3 ...

> # ---- 2. Sort by ID and merge (NOT cut-and-paste) ------------
> # all = TRUE is a FULL outer join: keeps an ID even if it only
> # appears in one  .... [TRUNCATED]

> dv <- dv[order(dv$ID), ]

> partA <- merge(iv, dv, by = "ID", all = TRUE)

> str(partA)
'data.frame':   569 obs. of  3 variables:
 $ ID: int  1 2 3 4 5 6 7 8 9 10 ...
 $ IV: num  3.37 3.93 6.3 5.71 5.48 ...
 $ DV: num  48.5 54 61.3 58.6 NA ...

> write.csv(partA, "data/partA_merged.csv", row.names = FALSE)

> # ---- 3. Counts the report asks for -------------------------
> n_total   <- nrow(partA)

> has_iv    <- !is.na(partA$IV)

> has_dv    <- !is.na(partA$DV)

> counts <- data.frame(
+   Quantity = c("Total subject IDs",
+                "IDs with an IV value",
+                "IDs with a DV value",
+       .... [TRUNCATED]

> print(counts)
                          Quantity    Value
1                Total subject IDs 569.0000
2             IDs with an IV value 500.0000
3              IDs with a DV value 520.0000
4          IDs with both IV and DV 454.0000
5 IDs with at least one of IV or DV 566.0000
6                  IDs missing both   3.0000
7           Fraction of IV missing   0.1213
8           Fraction of DV missing   0.0861

> md.pattern(partA)   # visual missing-data pattern (good for the appendix)
    ID DV IV
454  1  1  1   0
66   1  1  0   1
46   1  0  1   1
3    1  0  0   2
     0 49 69 118

> # ---- 4. Drop rows with NO information, then impute ----------
> partA_imp <- partA[has_iv | has_dv, ]
```

```
> # norm.boot = linear regression imputation with bootstrap.
> # Other allowed choices: "pmm" (predictive mean matching), "norm".
> # NOT allowed: mea .... [TRUNCATED]

> partA_complete <- cbind(ID = partA_imp$ID, complete(imp))

> md.pattern(partA_complete)   # should say "completely observed"
 /\     /\
{  `---'  }
{  O   O  }
==>  V <==  No need for mice. This data set is completely observed.
 \  \|/  /
  `-----'

    ID IV DV
566  1  1  1 0
     0  0  0 0

> write.csv(partA_complete, "data/partA_imputed.csv", row.names = FALSE)

> # ---- 5. Regression on the completed data --------------------
> M <- lm(DV ~ IV, data = partA_complete)

> summary(M)

Call:
lm(formula = DV ~ IV, data = partA_complete)

Residuals:
     Min       1Q   Median       3Q      Max
-10.5837  -1.8232   0.0143   2.0059   8.6393

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  30.6175     0.6607   46.34   <2e-16 ***
IV            4.8703     0.1288   37.80   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.019 on 564 degrees of freedom
Multiple R-squared:  0.717,     Adjusted R-squared:  0.7165
F-statistic:  1429 on 1 and 564 DF,  p-value: < 2.2e-16


> anova(M)
Analysis of Variance Table

Response: DV
           Df  Sum Sq Mean Sq F value    Pr(>F)
IV          1 13020.3 13020.3  1428.9 < 2.2e-16 ***
Residuals 564  5139.4     9.1
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

> confint(M, level = 0.95)
                2.5 %    97.5 %
(Intercept) 29.319672 31.915294
IV           4.617253  5.123397

> png("figures/partA_scatter.png", width = 700, height = 500)

> plot(DV ~ IV, data = partA_complete, pch = 20,
+      main = "Figure 1: DV vs IV (after imputation)")

> abline(M, col = "red", lwd = 2)

> dev.off()
RStudioGD
        2
```

File: output/partB_output.txt

```
> # ============================================================
> # AMS 315 - Project 1, Part B
> # Find a transformation, fit regression, bin x, la .... [TRUNCATED]
```

```
> dat <- read.csv("data/partB.csv", header = TRUE)

> str(dat)
'data.frame':   474 obs. of  3 variables:
 $ ID: int  1 2 3 4 5 6 7 8 9 10 ...
 $ x : num  1.98 1.64 1.61 1.16 1.69 ...
 $ y : num  0.0232 0.025 0.032 0.0311 0.0214 ...

> summary(dat)
       ID              x               y
 Min.   :  1.0   Min.   :1.008   Min.   :0.01574
 1st Qu.:119.2   1st Qu.:1.270   1st Qu.:0.02330
 Median :237.5   Median :1.530   Median :0.02761
 Mean   :237.5   Mean   :1.515   Mean   :0.02914
 3rd Qu.:355.8   3rd Qu.:1.759   3rd Qu.:0.03368
 Max.   :474.0   Max.   :2.000   Max.   :0.06684

> # ---- 1. Look at the raw data ---------------------------------
> M0 <- lm(y ~ x, data = dat)

> summary(M0)$r.squared            # baseline R-squared
[1] 0.5609992

> png("figures/partB_raw.png", width = 1000, height = 450)

> par(mfrow = c(1, 2))

> plot(y ~ x, data = dat, pch = 20, main = "Raw: y vs x"); abline(M0, col = "red")

> plot(fitted(M0), resid(M0), pch = 20, main = "Raw: residuals vs fitted"); abline(h = 0)

> dev.off()
RStudioGD
        2

> # ---- 2. Try candidate transformations (textbook 11.1) --------
> # Compare R-squared for each combination. Also LOOK at each
> # residual plot --  .... [TRUNCATED]

> ty <- list(y = function(v) v, `log(y)` = log, `1/y` = function(v) 1/v,
+            `sqrt(y)` = sqrt, `1/sqrt(y)` = function(v) 1/sqrt(v))

> r2 <- outer(names(tx), names(ty), Vectorize(function(a, b) {
+   summary(lm(ty[[b]](dat$y) ~ tx[[a]](dat$x)))$r.squared
+ }))

> dimnames(r2) <- list(names(tx), names(ty))

> round(r2, 4)
             y log(y)    1/y sqrt(y) 1/sqrt(y)
x       0.5610 0.6330 0.6602  0.6026    0.6520
log(x)  0.5626 0.6260 0.6434  0.6002    0.6401
1/x     0.5547 0.6087 0.6167  0.5877    0.6180
sqrt(x) 0.5630 0.6308 0.6530  0.6026    0.6473
x^2     0.5505 0.6302 0.6668  0.5956    0.6538

> # Box-Cox on y can suggest a power (lambda): -1 = 1/y, 0 = log(y), 0.5 = sqrt(y)
> # library(MASS); boxcox(M0)
>
> # ---- 3. Fit your chosen model  .... [TRUNCATED]

> dat$yt <- 1 / dat$y

> M1 <- lm(yt ~ xt, data = dat)

> summary(M1)

Call:
lm(formula = yt ~ xt, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-15.9499  -3.2655   0.4368   3.3128  15.7985

Coefficients:
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -1.4826     1.2779   -1.16    0.247
xt           25.1132     0.8293   30.28   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 5.107 on 472 degrees of freedom
Multiple R-squared:  0.6602,    Adjusted R-squared:  0.6595
F-statistic:   917 on 1 and 472 DF,  p-value: < 2.2e-16


> anova(M1)
Analysis of Variance Table

Response: yt
           Df Sum Sq Mean Sq F value    Pr(>F)
xt          1  23918 23917.7  916.98 < 2.2e-16 ***
Residuals 472  12311    26.1
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

> confint(M1, level = 0.95)
                2.5 %    97.5 %
(Intercept) -3.993664  1.028428
xt          23.483560 26.742787

> png("figures/partB_transformed.png", width = 1000, height = 450)

> par(mfrow = c(1, 2))

> plot(yt ~ xt, data = dat, pch = 20, main = "Transformed"); abline(M1, col = "red")

> plot(fitted(M1), resid(M1), pch = 20, main = "Transformed: residuals vs fitted"); abline(h = 0)

> dev.off()
RStudioGD
        2

> # ---- 4. Bin near-repeated x values --------------------------
> # Cut x into intervals; every x in an interval is replaced by the
> # interval's  .... [TRUNCATED]

> groups <- cut(dat$xt,
+               breaks = c(-Inf,
+                          seq(min(dat$xt) + width, max(dat$xt) - width, by = width),
+       .... [TRUNCATED]

> table(groups)
groups
(-Inf,1.06] (1.06,1.11] (1.11,1.16] (1.16,1.21] (1.21,1.26] (1.26,1.31]
         22          21          18          24          29          25
(1.31,1.36] (1.36,1.41] (1.41,1.46] (1.46,1.51] (1.51,1.56] (1.56,1.61]
         20          21          24          24          20          31
(1.61,1.66] (1.66,1.71] (1.71,1.76] (1.76,1.81] (1.81,1.86] (1.86,1.91]
         30          29          17          23          33          18
(1.91, Inf]
         45

> dat$x_bin <- ave(dat$xt, groups)       # mean x within each bin

> # ---- 5. Lack-of-fit test ------------------------------------
> fit_bin <- lm(yt ~ x_bin, data = dat)

> pureErrorAnova(fit_bin)
Analysis of Variance Table

Response: yt
             Df  Sum Sq Mean Sq  F value Pr(>F)
x_bin         1 23845.6 23845.6 919.9754 <2e-16 ***
Residuals   472 12383.3    26.2
 Lack of fit  17   589.8    34.7   1.3385 0.1639
 Pure Error  455 11793.5    25.9
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

> # Large p-value on the "Lack of fit" row => no evidence the straight
```

```
> # line is the wrong shape for the transformed data.
>
> # Optional: run the  .... [TRUNCATED]

> pureErrorAnova(lm(y ~ x_bin_raw, data = dat))
Analysis of Variance Table

Response: y
              Df    Sum Sq   Mean Sq F value Pr(>F)
x_bin_raw      1 0.0165439 0.0165439 596.346 <2e-16 ***
Residuals    472 0.0130575 0.0000277
 Lack of fit  17 0.0004348 0.0000256   0.922 0.5479
 Pure Error  455 0.0126226 0.0000277
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```